

Classification of Cancer Outcome

Clinical and Genomic Data Exploration

Team 5
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Context, Data, and Challenges

The Dataset:

- **Volume (n): 1231 patients** (Breast Cancer).
- **Data: Clinical** (23 variables: Age, stage, race, etc.) + **Genomic** (5000 genes).
- **Major Challenge: High Dimensionality** problem ($p = 5000 \gg n$).

Project Objective:

- **Target Variable:** `vital_status` (Vital Status).
- **Task:** Binary Classification (**Alive (0)** vs **Dead (1)**).

Challenge: Class Imbalance

- **84%:** Alive
- **16%:** Dead

Requires robust metrics (AUC, Sensitivity/Specificity) and an adapted threshold.

Variable Types

- **Numerical (5):** Age, follow-up days, weight...
- **Categorical (18):** Race, tumor, treatment...

Data Cleaning and Feature Engineering

Cleaning and Removals

- **Data Leakage:** Removal of `follow_ups` (post-treatment).
- **Constants:** Removal of `disease_type` and `primary_site`.
- **Low Variance:** Removal of `tissue_or_organ_of_origin`, `gender`, `ethnicity`, and `age_is_obfuscated`.
- **Uniqueness:** Removal of `patient_barcode`.
- **Missing Target:** Removal of patients without `vital_status`.
- **Missing Observations:** Removal of rows with more than 30% missing data.

Feature Engineering

- **NA Standardization:** "Unknown", "Not Reported", "NULL" → NA.
- **Median Imputation:** Applied to missing numerical values.
- **Anatomical Sites:** Simplification into **binary variable** (Standard Mammary (0) vs Atypical (1)).
- **Histology:** Grouping of `primary_diagnosis` into 3 classes (Ductal / Lobular / Other).
- **Tumor Stage (AJCC):** Grouping of substages (T1a, T1b → T1 / Tis, TX, NA → Other_Unknown).
- **Tumor Type:** Creation of a binary variable (Primary vs Non-Primary).

Threshold Selection

With 84% "Alive", Accuracy is misleading. We must find a custom threshold to avoid missing deaths.

The Principle

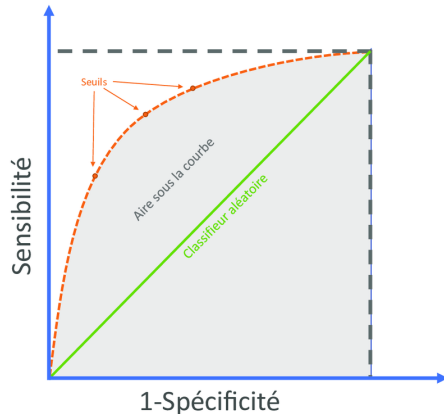
The ROC curve plots performance for **all possible thresholds** (from 0 to 1).

- It does not decide, it proposes.

Youden's J Statistic

This is the mathematical method to choose the **best point** on the curve:

$$J = \max(\text{Sensitivity} + \text{Specificity} - 1)$$



Baseline Models

Clinical

A. Raw Clinical (Biased)

Result: AUC 0.7204, Acc 84.7%

- **Problem:** Sensitivity 33.3%.
- **Diagnosis:** Model bias towards the majority class ("Alive").

B. Imputed Clinical (Corrected)

Result: AUC 0.8309, Acc 73.8%

- **Gain:** Sensitivity 81.6%.
- **Diagnosis:** Imputation corrects bias and recovers the clinical signal. This is the reference.

Genomic

C. Naive GeneX (Naive Failure)

Result: AUC 0.5588, Acc 55.3%

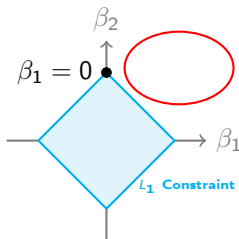
- **Problem:** Noise dominates the signal.
- **Diagnosis:** The model flips a coin. The $p \gg n$ brute force approach is impossible.

Baselines Summary

Verdict:

- The **Clinical data (imputed)** is the best baseline (AUC 0.83).
- **Genomic data** imperatively requires a **dimension reduction (Screening)** step before any modeling.

Lasso on GeneX



Lasso Performance (Genes only)

Selected Genes	41 / 5000
Accuracy	73.0%
Sensitivity	31.6%
AUC	0.64

The Problem: Dimensionality

With $p = 5000$ genes and $n = 1200$ patients, classic regression is impossible (dominant noise).

The Geometric Solution

The "pointed" constraint (the blue diamond) forces the solution into the corners.

- **Effect:** Strict zeroing of 4959 coefficients (noise).
- **Observation:** The model cleans the data well (Acc > 50)

Hybrid Strategies: Clinical + Genomic Integration

A. Global Lasso (Merged)

All mixed: Clinical + 5000 Genes.

Accuracy: 75.1% Sensitivity: 55.3%

	Pred 0	Pred 1
Actual 0	157	17
Actual 1	42	21

B. Lasso Screening → GLM

Selection (Lasso) → Final GLM.

Accuracy: 79.8% Sensitivity: 54.4%

	Pred 0	Pred 1
Actual 0	164	21
Actual 1	27	25

Methodological Verdict

Approach **B (Two-Step)** is superior (+4.6% Accuracy).

- It uses Lasso only to reduce dimensionality (5000 → 41).
- It lets the GLM optimize final predictions without biasing clinical variables.

AIC Selection

Failure of Brute Force Approach (Stepwise on 5000 genes)

Diagnosis: The algorithm runs indefinitely.

- **Cause:** Combinatorial Explosion ($p = 5000$) $\implies$ search space too vast.
- **Verdict:** *A dimension reduction step is indispensable.*

Stepwise on Top 50 (via T-test)

Key Metrics (Test)

- **AUC:** 0.7560
- **Accuracy:** 70.04%
- **Sensitivity:** 68.42% (Significant gain vs 57% naive)
- **Specificity:** 70.35%

Confusion Matrix

	Pred 0	Pred 1
Actual 0	140	59
Actual 1	12	26

Key Analysis

- **Feasibility:** Reduction (T-test $\rightarrow$ Top 50) unlocks AIC convergence.
- **Trade-off:** Sensitivity is good (68.4%), but the model is ****too risky**** (59 False Positives).

Fisher

Key Metrics (Test Set)

- Accuracy (Global Precision): 0.7342
- Specificity (Class 0): 0.7487
- Sensitivity (Class 1): 0.6739
- AUC (Area Under the Curve): **0.7666**

Analysis: Improvement vs T-test

The Fisher Score (0.7666) is superior to Stepwise after T-test (0.7560).

- Better global performance (AUC).
- Better control of False Positives (FP divided by 4).

Fisher's method is therefore retained as the best pre-screening.

Confusion Matrix

		Prediction	
		Deceased (1)	Alive (0)
Actual	Deceased (1)	31 (TP)	48 (FN)
	Alive (0)	15 (FP)	143 (TN)

Analysis Detail: The false positive rate (FP=15) is significantly reduced (vs 59 previously), improving Specificity and reliability.

F-Score Principle

$$F(i) = \frac{(\mu_{i,1} - \mu_{i,0})^2}{\sigma_{i,1}^2 + \sigma_{i,0}^2}$$

μ and σ^2 are the mean and variance of gene i expression in each class.

Random Forest: Theory and Interpretation

Theory: Bagging on Selected Variables

- **Principle:** Majority vote of 500 independent trees (Bagging).
- **Objective:** Capture **non-linear interactions** on the 41 Lasso genes.

Results (Test)

- Acc: **83.5%**
- Spec: **97.9%**
- Sens: **23.9%**

	P. Alive	P. Dead
A. Alive	187	35
A. Dead	4	11

Why this failure on sensitivity?

- RF seeks to minimize global error.
- **The Trap:** With only 16% deaths, the model "learns" that it is safer to predict "Alive".
- **Effect:** It becomes **conservative**. It takes no risks (few False Positives) but misses the target (many False Negatives).

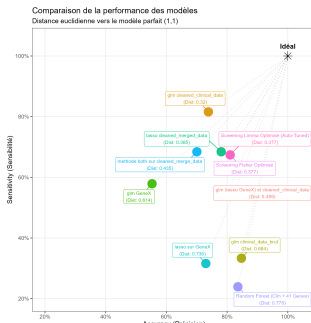
Verdict: Complexity does not pay off

The non-linear approach fails to handle class imbalance. The weighted linear approach (Lasso → GLM) remains superior for screening (Sensitivity 54%).

Results Synthesis and Final Model Choice

Comparative Table (Test Set)

Model	AUC	Acc	Sens.	Spec.
Part A: References				
Raw Clinical (Biased)	0.720	84.7%	33.3%	93.0%
Clinical (Imputed)	0.831	73.8%	81.6%	72.4%
3. GLM GeneX (Standard)	0.559	55.3%	57.9%	N/A
Part B: Advanced Models				
4. Lasso GeneX	0.637	73.0%	31.6%	80.9%
5. Random Forest	0.733	83.5%	23.9%	97.9%
6. Stepwise (Top 50)	0.756	70.0%	68.4%	70.4%
7. Fisher Screening + Clin	0.767	73.4%	67.4%	74.9%
9. Combined Lasso (Merged)	0.769	78.1%	68.4%	79.9%
10. GLM (Lasso GeneX) + Clin	0.771	76.0%	58.7%	80.1%



Clinical Model (Baseline)

The Imputed Clinical Model (AUC 0.831)

- Best AUC and Sensitivity (81.6%) for initial screening.
- Serves as reference for rapid diagnosis and follow-up.

Genomic Model (Refinement)

Fisher Screening + Clinical (AUC 0.767)

- Best performance/simplicity trade-off for gene-based models.
- Minimum distance to ideal among genomic models.

Deployment Strategy

2-Step Strategy:

- The Doctor uses the Clinical Baseline for rapid screening.
- The Fisher model is used as a second line to refine Precision (Accuracy) and interpretation when genetic data is available.

Limitations and Perspectives

Model Limitations

Class Imbalance (84% vs 16%)

- Low number of deaths $\implies$ tendency to favor the majority class.
- Impact: Drop in **Sensitivity** (risk of False Negatives).

Data Complexity

- Complex cleaning (heterogeneous variables) may have introduced **statistical noise**.
- Dilution of relevant genomic information.

Future Work and Perspectives

Biological Approach (Pathway)

- Context: Treat genes by **Signaling Pathways** (e.g., KEGG, Reactome).
- Objective: **Intelligent reduction** (5000 $\rightarrow$ $\sim$ 200 scores) and **Robustness** to noise.

Capturing Non-Linearity

- Leads: Boosting Models (**XGBoost**, LightGBM) or SVM.
- Objective: Capture **cross-interactions** and complex relationships missed by GLM/Lasso.